

$$r = 0.50 \pm 0.01 \text{ cm}$$

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Ans: D

$$v_x = u_x$$

$$v_y^2 = u_y^2 - 2gs$$

$$= (50\sin 30^\circ)^2 + 2(-9.81)(-30)$$

$$v_y = -34.8 \text{ m s}^{-1}$$

The negative sign shows that the ball is moving downwards.

$$\theta = \tan^{-1} (v_x / v_y) = (50\cos 30^\circ / 34.8)$$

$$= 51.2^\circ$$

